Using Kernel Density Function and Student's t-distribution to model Value at Risk and Expected Shortfall using Monte Carlo method

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***Abstract*—Basel III regulation imposes a clear market risk frameworks for financial institutions. The principal shift rests in the core of the attention now turned toward a 97.5% Expected Shortfall, rather than VaR at 99% via FRTB.**

In this paper we use Monte Carlo simulations through Student's t-distribution and non-parametric Kernel Density Estimator (KDE) to calculate VaR and CVaR at the 99% and 97.5% threshold.

Empirically, our simulations demonstrate that the Geometric Brownian Motion framework penalizes risk architecture by collapsing the threshold difference between Value at Risk and Expected Shortfall. In contrast, both the parametric Student's t-distribution and the non-parametric Kernel Density Estimator successfully capture extreme market shocks with fatter tail distributions. In each scenario, we observe that $CVaR(97.5\%)$ is greater than $VaR(99\%)$, validating the Basel III Fundamental Review of the Trading Book (FRTB) guidelines.

We demonstrate empirically that KDE captures the asymmetric tail structure of real returns that Student's t cannot, with implications for Basel III capital adequacy.

***Index Terms*—Value at Risk, Expected Shortfall, Monte Carlo modeling, Kernel Density Estimator, Student's t-distribution, Asymmetric tail risk**

I. Introduction

Value at Risk is defined as: "a measure of the worst expected loss on a portfolio of instruments resulting from market movements over a given time horizon and a pre-defined confidence level" [1] [2]. Formally, it can be written as follow:

$$VaR_\alpha = \inf\{l \in \mathbb{R} : \mathbb{P}(L > l) \leq 1 - \alpha\}$$

Expected Shortfall or Conditional Value at Risk is defined as: "a measure of the average of all potential losses exceeding the VaR at a given confidence level" [1]. Mathematically, we can write it as:

$$CVaR_\alpha = \mathbb{E}[L|L \geq VaR_\alpha]$$

Under the Basel III regulation, the confidence level for VaR is $\alpha = 99\%$, even though it focuses primarily on Expected Shortfall. Developed by the Basel Committee on Banking Supervisions (BCBS), this regulation was first implemented in July 2013 and finally reformed in December 2017. Its first aim was to respond to the 2007-2008 global financial crisis. To counter these risks it mainly focuses on enhanced capital requirements, non-risk-based leverage ratio, and counterparty credit risk.

In terms of market risk management, Basel III —specifically through the Fundamental Review of the Trading Book (FRTB)— shifts the core metric from a 99% Value at Risk (VaR) to a 97.5% Expected Shortfall (ES). This shift implies that regulators no longer satisfied with knowing just the threshold of a bank's potential trading losses, but rather the actual severity of losses when a tail-risk event occurs. Additionally, while the previous VaR framework assumed a static, short-term liquidity horizon across all asset classes, the new CVaR-based framework forces banks to map different liquidity horizons to different risk factors.

Banks aren't talkative on the subject and it's hard to really understand, when they are using the Monte Carlo method, which diffusion model they use [3]. Our goal here will be to compare three different approaches. First, we will use a Geometric Brownian motion, that assumes the logarithm of asset prices follows a Brownian motion with a drift. It implies that the probability of extreme events is highly underestimated. In order to offset that, a Student's t-distribution will be implemented to take into account fat-tail risks. This model is usually a better fit for financial returns and it captures extreme events more accurately. Finally, we are going to use a Kernel Density Estimator which is a non-parametric way to estimate the true distribution of financial returns directly from data.

II. Usual methods

Parametric method

Generally, Basel III allows for three methods: parametric, historic, and Monte Carlo.

Parametric method simply assumes loss follows a Gaussian (normal) distribution, then we have: $L \sim N(\mu_L, \sigma_L^2) \Rightarrow L = \mu_L + \sigma_L U$, with $U \sim N(0, 1)$.

Under this method, the formulas are:

$$VaR_\alpha = \mu_L + \sigma_L \Phi^{-1}(\alpha)$$

Applying this to our definition of Expected Shortfall, it implies:

$$CVaR_\alpha = \mathbb{E}[L|L \geq VaR_\alpha]$$

$$= \frac{1}{1-\alpha} \int_{VaR_\alpha}^{+\infty} l f_L(l) dl \mu_L + \sigma_L \frac{\varphi(\Phi^{-1}(\alpha))}{1-\alpha}$$

(See Appendix 1. for demonstrations.)

Historical method

This method rests on the assumption that the worst case scenarios of the period taken into account will remain the same. Nevertheless, reading [3], we can see that the distribution can still be adapted following some scenarios. Furthermore, it assumes portfolio remains the same, the weight of each asset being unchanged.

Let $r_{i,t}$ be the daily returns for each asset i at date $t \in [i, T]$ our historical interval. We apply today's weight w to each historical return scenario:

$L_t = -\sum_{i=1}^{T} \omega_i r_{i,t} = -w^T r_t$ (it takes the opposite value since loss is considered the opposite of the asset return). It gives us T realized loss scenarios, using the current portfolio composition replayed through history. We order the T losses in ascending order: $L_{(1)} < L_{(2)} < ... < L_{(T)}$.

Finally, we have:

$$VaR_\alpha = L_{[\alpha,T]}$$

$$CVaR_\alpha = \frac{1}{T - \alpha T} \sum_{t=\alpha t+1}^{T} L_{(t)}$$

III. Monte Carlo Method

Method framework

The Monte Carlo method is a way to find answers to hard financial problems by using random numbers and simulating thousands of independent paths following the same distribution. First, we choose a statistical rule for how our assets behave (for instance a log-normal distribution or a Student's t-distribution). It allows us to see where the stock prices can potential go. Finally, we put all these simulated results together from the worst loss to the biggest profit. This gives us a huge list of possibilities, and we just look at the bottom 1% of the worst results to easily find the Value at Risk (VaR) or take the average of those bad results to get the Conditional VaR (CVaR).

Statistical distributions

Our goal now is to implement and compare three different statistical distributions: log-normal, Student's t, and Kernel Density Estimator.

Log-normal distribution

Geometric Brownian motion makes sure prices are strictly positive by using an exponential. Also, it scales volatility and drift based on the current price. Finally, this model assumes that continuously compounded returns are normally distributed. If we assume that an asset S follows a log-normal distribution, we have:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Where S_t is the asset price at time t, μ the drift (expected return), σ the volatility coefficient, dt an infinitesimal time step, dW_t a Wiener process (or Brownian motion) increment, which captures the random market shocks where $dW_T \sim N(0, dt)$ [4]. It has the following properties:

1) $W_0 = 0$.
2) It has independent increments.
3) For any two time steps s and t (where $t > s$), the change in value is normally distributed: $W_t - W_s \sim N(0, t-s)$.

The analytical solution, by applying Itô's Lemma [5] to solve the SDE, gives:

$$S_T = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) T + \sigma W_T\right)$$

(Demonstrations are shown in Appendix 2.)

If we set $s = 0$ and look strictly at a single future maturity date $t = T$, the total increment is:

$$W_T - W_0 = W_T \sim N(0, T)$$

If we scale a standard normal random variable $Z \sim N(0, 1)$ by multiplying it by $\sqrt{T}$, its mean remains 0 and its variance becomes $(\sqrt{T})^2 = T$. Therefore:

$$W_T \overset{d}{=} Z\sqrt{T}$$

Since W_T can be written as $Z\sqrt{T}$, where $Z \sim N(0, 1)$, the formula used is:

$$S_T = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) T + \sigma Z\sqrt{T}\right)$$

Student's T-distribution [6]

The Student's t-distribution does not only rely on the mean and variance, but introduces a third parameter: degrees of freedom (ν). Further in this paper, we use $\nu = 4$ as a conservative choice to ensure fat-tailed behavior while maintaining finite variance. The mathematical formula for the Probability Density Function (PDF) of such a distribution is:

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Where the Gamma function is defined as:

$$\Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt$$

If the degrees of freedom is low ν the distribution has very fat tails and a higher peak (leptokurtic). However when $\nu \to \infty$ the distribution mathematically converges into the standard normal distribution.

Modeling an asset using this distribution, we get:

$$S_T = S_0 \exp\left(\mu T + \sigma \sqrt{\frac{\nu - 2}{\nu}} T_\nu \sqrt{T}\right)$$

Where T_ν is a random variable drawn from a standard Student's t-distribution with ν degrees of freedom and $\sqrt{\frac{\nu-2}{\nu}}$ is the scaling factor ensuring that the variance of the return over time matches our target volatility. (Proof is given in Appendix 3.)

Kernel Density Estimator

The formal mathematical formula for a Kernel Density Estimator is:

$$\hat{f}_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

Where h is the bandwidth , and $K(\cdot)$ the Kernel function. If we use the Gaussian (Normal) Kernel: $K(u) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}u^2}$, we finally get:

$$\hat{f}_h(x)\frac{1}{nh\sqrt{2\pi}}\sum_{i=1}^{n} e^{-\frac{1}{2}(\frac{x-x_i}{h})^2}$$

the summation gives us exactly the probability density function (PDF) of a normal distribution $N(x_i, h^2)$. Therefore, the KDE can be written as:

$$\hat{f}_h(x) = \frac{1}{n}\sum_{i=1}^{n} f_N(x|\mu = x_i, \sigma^2 = h^2)$$

This tells us that the KDE treats the asset return distribution as a uniform mixture of n distinct normal distributions. Assuming that $y_j \sim N(x_i, h^2)$ we have that:

$$y_j = x_i + hZ_j$$

To decide which value to choose for h we will use the Silverman's Rule of Thumb [7], that states:

$$h = \left(\frac{4\hat{\sigma}^5}{3n}\right)^{\frac{1}{5}} \approx 1.05\hat{\sigma}n^{-\frac{1}{5}}$$

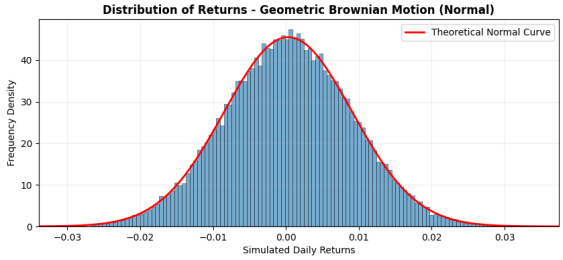
The core difference of this approach compared to a Brownian motion is that it assume no prior functional form for the distribution. Instead of forcing the data into a bell curve, the data defines its own shape.

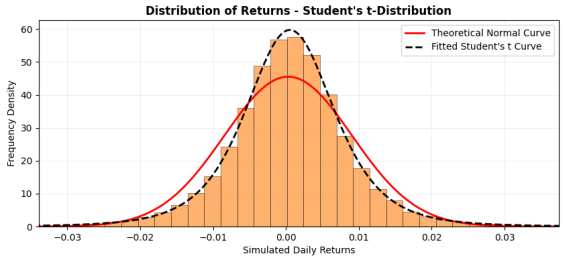
How to decide which distribution model to use?

KDE (Non-Parametric) does not assume any underlying shape for our data. It builds a distribution directly from historical data points by placing a "kernel" (smooth curve) over each observation. It is purely empirical. Student's t (Parametric) assumes a fixed mathematical shape defined by specific parameters (mean, scale, and degrees of freedom). It is symmetrical but features heavy tails, making it excellent for capturing extreme financial shocks. GBM (Stochastic Process) unlike the other two, GBM is not a static probability distribution—it is a time-dependent path generator. It models how an asset price moves over time based on a constant drift (trend) and volatility (random walk).

IV. Implementations & Results

Using Python, we can simulate an asset N times and adjust the characterizing parameters as required for our study. The python code can be found in Appendix 4. For our study, we construct an equally-weighted portfolio made of: S&P 500, NASDAQ, CAC40, FTSE 100, DAX, and Hang Seng. This portfolio is of course a simplified version of what could be seen in practice but we assume for a CVaR and VaR simulation it can fit. We extract the data for a five years period, from 06/01/2021 to 06/01/2026, with yfinance module in Python. The prices used are daily close prices and we consider log-returns since it assumes symmetry around zero which is useful for modeling. Mean and standard deviation are computed via a Maximum Likelihood estimation which provides a uniform, consistent framework to compare the normal distribution against non-Gaussian models (like the Student's t-distribution) that require statistical fitting rather than simple averages. After gaining these data, we model 50,000 paths for each distribution. This presents some assumptions and simplifications. In fact, real trading books re-balance often which is not the case here in our model. To get a better picture of how returns are distributed, here are the histograms representing the distribution of our generated-data.





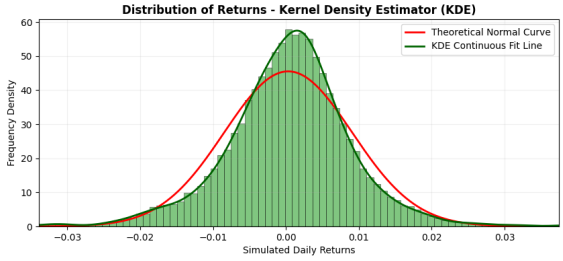
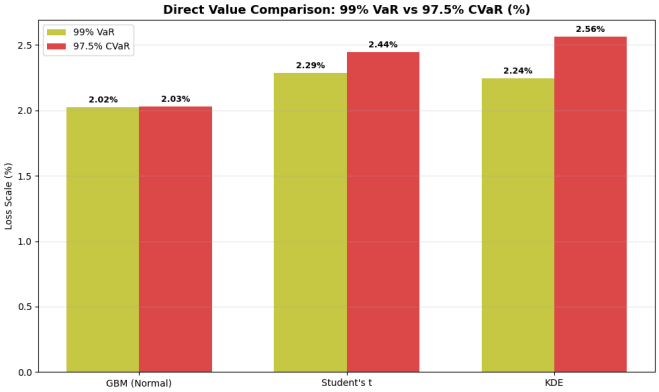


TABLE I: Statistical Significance Tests for Skewness and Kurtosis

Model	Test / Metric	Value / Stat	z-score	p-value
GBM	Skewness	-0.0095	-0.8703	0.384119
	Ex. Kurtosis	+0.0010	+0.0465	0.962946
Student's t	Skewness	+0.0004	+0.0330	0.973636
	Ex. Kurtosis	+14.7932	+675.2208	0.000000***
KDE	Skewness	-1.0377	-94.7309	0.000000***
	Ex. Kurtosis	+11.2080	+511.5783	0.000000***

Significance codes: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$



The Geometric Brownian Motion (GBM) model assumes that asset returns follow a perfect normal distribution. In our simulation, the GBM skewness yields a z-score of -0.8703 ($p = 0.384$), and the excess kurtosis yields a z-score of $+0.0465$ ($p = 0.963$). Because both p-values are much larger than 0.05, these results are not statistically significant, meaning the distribution is perfectly Gaussian. Consequently, the GBM model is completely blind to unexpected market shocks or severe tail risks, making it an unrealistic tool for downside risk management.

To fix the limitations of the normal model, a Student's t-distribution was fitted to the data. While the skewness remains statistically insignificant ($z = +0.0330$, $p = 0.974$), the excess kurtosis shifts drastically, showing an extreme z-score of $+675.2208$ ($p < 0.001$). This highly significant p-value proves that the asset's returns have much fatter tails than a normal distribution. On the chart, the Student's t curve successfully captures the heavy risk in the tails, making it a much more reliable model for calculating portfolio Value-at-Risk (VaR).

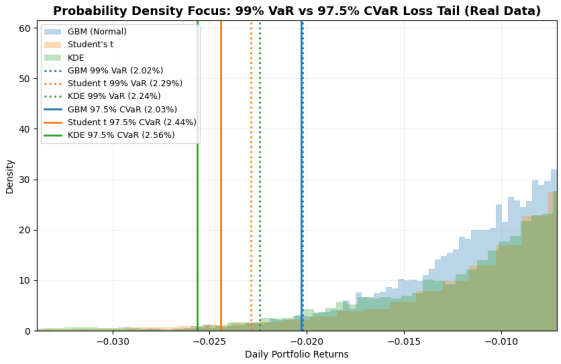
Although the Student's t-distribution models fat tails well, its mathematical structure forces the data to be perfectly symmetric. The non-parametric Kernel Density Estimator (KDE) removes this restriction and reveals a highly significant negative skewness with a z-score of -94.7309 ($p < 0.001$), alongside a significant excess kurtosis z-score of $+511.5783$ ($p < 0.001$). These significant p-values confirm that the distribution is heavily distorted to the left. This proves the true asymmetry of financial risk: asset returns typically face sudden, severe negative crashes rather than positive jumps.

Finally, putting altogether our estimated risk measures:

This histogram shows a big difference between the models. Under the normal GBM model, the $VaR(99\%) = 2.02\%$ and the $CVaR(97.5\%) = 2.03\%$ are almost exactly the same, with a gap of only 0.01%. This happens because the normal model assumes that extreme losses are nearly impossible. In the real world, using this normal model is dangerous because it leaves a portfolio unprotected against big market drops.

When we look at the other models, the risk numbers increase to show the true dangers of the market. The Student's t-distribution gives a $VaR(99\%)$ of 2.29% and a $CVaR(97.5\%)$ of 2.44%, showing a larger gap of 0.15% because it captures fat tails. However, the KDE model gives the most realistic view of risk. Even though its $VaR(99\%)$ is 2.24%, its $CVaR(97.5\%)$ jumps to the highest level at 2.56%. This 0.32% gap is caused by the negative skewness of the data, proving that real-world market crashes are much worse than simple models predict.

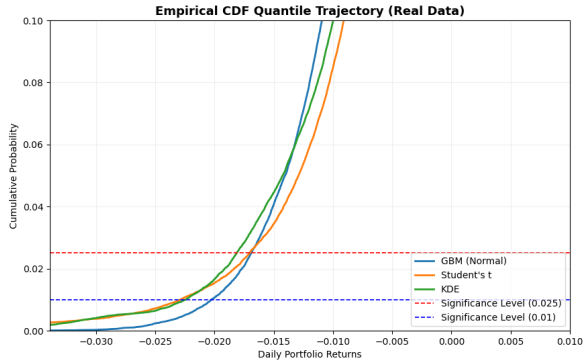
Also, we notice that in every case CVaR(97.5%) is above VaR(99%) reflecting the reinforcement in capital exposition regulation imposed by Basel III.



The graph of Probability Density shows the worst-case losses for the three different models. The normal model (GBM) is represented by the blue lines. Here, the $VaR(99\%)$ is 2.02% and the $CVaR(97.5\%)$ is 2.03%. Because these two lines are almost on top of each other, the model assumes that

there are no risk of extreme scenarios.
The other two models look at the actual danger in the market much better. As seen in the graph, the Student's t distribution (orange lines) separates the VaR (2.29%) and CVaR (2.44%) because it takes into consideration the market has fat tails. However, the KDE model (green lines) shows the most realistic danger. Even though its $VaR(99\%)$ is 2.24%, its $CVaR(97.5\%)$ moves way to the left at 2.56%. This large gap happens because real financial data has a negative skew, meaning that when markets crash, they fall fast and deep.

CVaR(97.5%) is always represented at the left (deeper into high-loss territory). Unlike the normal distribution, KDE and Student's t-histogram bars continue to exists on the left part of the graph. This illustrates again the reason why Basel III guidelines transitioned regulatory capital calculations to a CVaR(97.5%) framework.



The chart empirical CDF shows the cumulative probability of negative return. The two horizontal dashed lines represent the risk thresholds: the red line is the 2.5% risk level (used for 97.5% CVaR), and the blue line is the 1% risk level (used for 99% VaR). This chart shows exactly where each model intersects these risk lines.
As we look more closely, we can see how the lines cross the thresholds differently.

- The blue GBM (blue) line stays to the right of the others.
- However, both the Student's t (orange) and KDE (green) lines are much further to the left.

The fact that the orange and green curves sit higher up on the left side proves that large losses happen much more often in real life than the normal GBM model expects.

V. Limitations

Our entire simulation operates on a one-day return horizon. FRTB explicitly requires liquidity-adjusted horizons ranging from 10 days to 120 days. Our results therefore represent a best-case, most liquid scenario and would need to be scaled before being operationally used in a real regulatory context.

Our Monte Carlo draws each scenario as if the six indices move independently. In reality, tail dependence between equity markets is well documented: during the 2020 Covid crash or 2008 crisis, global indices fell simultaneously and correlations spiked toward 1. A copula-based approach would capture this.

The equal-weight assumption is held fixed throughout the whole time-period where we collect data. In reality, portfolio allocation evolves continuously which changes the loss distribution.

Silverman's rule-of-thumb is considered optimal for estimating the bulk of a distribution, but also it is known to oversmooth the tails. A smaller bandwith would produce heavier tails and at the same time higher risk measures estimates.

The Student's t-distribution is symmetric by construction, meaning it assigns equal probability to extreme gains and extreme losses. Since financial returns exhibit asymmetric crash risk (negative skewness), the Student's systematically underestimates left-tail severity relative to a model that allows for skewness.

VI. Backtesting

Using data from 2015 to 2024, we can split the sample between training period and test period. Here we divide it from 2015 to 2020 and the second part is from 2021 to 2024. Here are the results:

TABLE II: VaR/CVaR Estimates (fitted on 2015–2020)

Model	**VaR(99%)**	**CVaR(97.5%)**
GBM	2.43%	2.43%
Student t	2.89%	3.36%
KDE	3.21%	3.63%

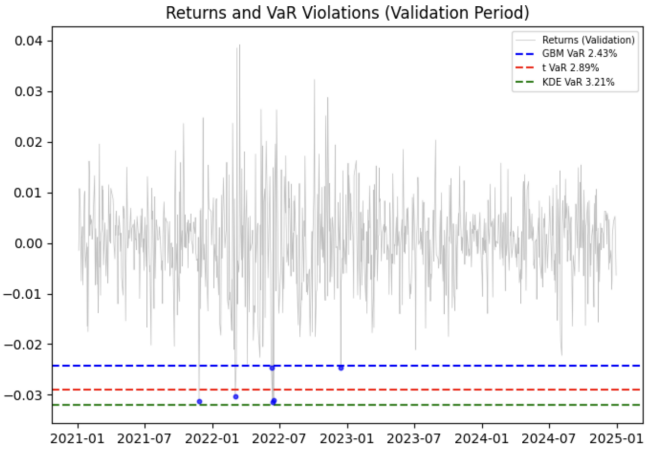
TABLE III: Backtest: VaR(99%) Violation Rates (2021–2024)
Theoretical target: 1.00% violation rate

Model	**Violation Rate**	**Judgment**
GBM	0.64%	lax
Student t	0.43%	slightly conservative
KDE	0.00%	slightly conservative

We again observe that GBM underestimates tail risks, both measures being equal. Half of the black swan events result in a breach. KDE's 0.00% violation rate suggests a slightly conservatism in practice. A 0% violation rate means excess capital allocation, which is inefficient from a regulatory standpoint. Student's t-distribution appears most balanced between accuracy and conservatism. Black swan events (COVID: -7.58%) exceeded all three models — no parametric or non-parametric approach can fully capture them. However, KDE approach is the most robust approach to extreme scenarios like these ones.

TABLE IV: Key Event Analysis

Event	Return	GBM	Student t	KDE
COVID crash (2020/3)	-7.58%	Breach	Breach	Breach
Rate hike shock (2022/6)	-3.15%	Breach	Breach	OK
Russia invasion (2022/2)	-1.73%	OK	OK	OK
SVB collapse (2023/3)	-1.07%	OK	OK	OK



VII. Conclusion

This paper introduces three distributions to evaluate Value at Risk and Expected Shortfall under the Basel III/FRTB regulation. We modeled 50,000 different random paths for each distribution in order to evaluate these risk measures.
The first distribution used is a geometric Brownian motion (using Gaussian distribution assumptions), and our findings show that it fails to capture fat-tail risk. Indeed, as calculated, the excess kurtosis is near zero showing the lack of consideration for extreme scenarios.
The second distribution implemented is the Student's t-distribution in order to overcome that issue of not considering fat-tail risk. This distribution exhibits a clearly positive excess kurtosis which implies a leptokurtic distribution as we can observe with the corresponding histogram. Finally, the Kernel Density Estimator also exhibits a leptokurtic distribution and a clearly lower skewness than the two others. As the data tells, this distribution gives the highest Expected Shortfall but still has a lower Value at Risk, relative to the thresholds imposed by the regulation. This model is different in the fact that it is non-parametric, it let the data defines it own form.

Our findings demonstrate the danger of normality assumption. As we showed, the Gaussian distribution does not take consideration of fat-tail risks that are essential in risk measurement. Backtesting helps at seeing the weakness of this distribution since it is the one with the highest violation rate. On the one side, data clearly shows the geometric Brownian motion exhibits lower values for both measures (Expected Shortfall and Value at Risk) and gives almost the value for both of them showing again the ignorance of extreme scenarios. On the other side, KDE as well as Student's t distributions have a much higher CVaR(97.5%) than VaR(99%) proving the utility of the Basel III regulation that shifts toward the first one in order to reinforce capital requirement.

A parametric approach assumes that the distribution of a portfolio returns follows a known statistical distribution equation. To capture fat tails with a parametric model, Gaussian distribution must be forgotten. In a non-parametric framework like the Kernel Density Estimator (KDE), we get rid of equations to use data directly. Then, it captures leptokurtosis naturally. This gives more accurate measures, especially for hidden structural risks (such as the negative skewness caused by market sell-offs).

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VIII. Appendices

Appendix 1. — demonstrations for parametric method

VaR is such that [2]:

$$\mathbb{P}(L < VaR_\alpha) = \alpha$$

Since we assumed $L \sim N(\mu_L, \sigma_L)$:

$$\Leftrightarrow \mathbb{P}(\mu_L + \sigma_L U < VaR_\alpha) = \alpha$$

With $U \sim N(0,1)$, We can simplify as:

$$\Leftrightarrow \mathbb{P}\left(U < \frac{VaR_\alpha - \mu_L}{\sigma_L}\right) = \alpha$$

$$\Leftrightarrow \Phi\left(\frac{VaR_\alpha - \mu_L}{\sigma_L}\right) = \alpha$$

$$\Leftrightarrow \frac{VaR_\alpha - \mu_L}{\sigma_L} = \Phi^{-1}(\alpha) \Leftrightarrow VaR_\alpha = \mu_L + \sigma_L \Phi^{-1}(\alpha)$$

Applying this to our definition of Expected Shortfall, it implies:

$$CVaR_\alpha = \mathbb{E}[L|L \geq VaR_\alpha] = \frac{1}{1-\alpha}\int_{VaR_\alpha}^{+\infty} l f_L(l) dl$$

with f_L the probability density function of $L \sim N(\mu_L, \sigma_L^2)$. Hence:

$$f_L(l) = \frac{1}{\sqrt{2\pi\sigma_L^2}} \exp\left(-\frac{(l-\mu_L)^2}{2\sigma_L^2}\right)$$

Let

$$\varphi : \frac{l-\mu_L}{\sigma_L} \to \frac{1}{\sqrt{2\pi}} \exp\left(-\left(\frac{l-\mu_L}{\sigma_L^2}\right)^2 /2\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-(l-\mu_L)^2}{2\sigma_L^2}\right)$$

This gives us:

$$f_L(l) = \frac{1}{\sigma_L}\varphi\left(\frac{l-\mu_L}{\sigma_L}\right)$$

Finally, we can compute the Expected Shortfall:

$$CVaR_\alpha(L) = \frac{1}{1-\alpha}\int_{VaR_\alpha(L)}^{+\infty} \frac{l}{\sigma_L}\varphi\left(\frac{l-\mu_L}{\sigma_L}\right) dl$$

Let's define:

$$z = \frac{l-\mu_L}{\sigma_L} \Leftrightarrow l = \mu_L + \sigma_L z \Rightarrow dl = \sigma_L dz$$

This gives us:

$$\frac{1}{\sigma_L}\varphi\left(\frac{l-\mu_L}{\sigma_L}\right)\sigma_L dz = \varphi(z)dz$$

$$VaR_\alpha = \mu_L + \sigma_L \Phi^{-1}(\alpha)$$

$$z = \frac{l-\mu_L}{\sigma_L} = \frac{(\mu_L + \sigma_L\Phi^{-1}(\alpha) - \mu_L)}{\sigma_L} = \Phi^{-1}(\alpha)$$

Finally:

$$CVaR_\alpha(L) = \frac{1}{1-\alpha}\int_{\Phi^{-1}(\alpha)}^{+\infty} (\mu_L + \sigma_L z)\varphi(z)dz$$

$$CVaR_\alpha = \frac{1}{1-\alpha}\left(\mu_L\int_{\Phi^{-1}(\alpha)}^{+\infty}\varphi(z)dz + \sigma_L\int_{\Phi^{-1}(\alpha)}^{+\infty} z\varphi(z)dz\right)$$

Let's compute separately these two integrals:

$$I_1 = \int_{\Phi^{-1}(\alpha)}^{+\infty}\varphi(z)dz = 1 - \Phi(\Phi^{-1}(\alpha)) = 1-\alpha$$

$$I_2 = \int_{\Phi^{-1}(\alpha)}^{+\infty} z\varphi(z)dz$$

$$= -\int_{\Phi^{-1}(\alpha)}^{+\infty}\varphi'(z)dz = -\Big[\varphi(z)\Big]_{\Phi^{-1}(\alpha)}^{+\infty} = \varphi(\Phi^{-1}(\alpha))$$

We use here two arguments:

- $\varphi'(z) = -z\varphi(z)$
- $\varphi(z) \to 0$ when $z \to +\infty$

Hence:

$$CVaR_\alpha = \frac{1}{1-\alpha}\left[\mu_L(1-\alpha) + \sigma_L\varphi(\Phi^{-1}(\alpha))\right]$$

$$\Leftrightarrow CVaR_\alpha = \mu_L + \sigma_L\frac{\varphi(\Phi^{-1}(\alpha))}{1-\alpha}$$

Appendix 2. — Brownian motion

We let the following formula:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Let: $f(S_t) = ln(S_t)$. For a continuous function $f(t, S_t)$ the second-order Taylor expansion using Itô's Lemma [5] gives us:

$$df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S}dS_t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}(dS_t)^2$$

Let's calculate the partial derivatives for our specific function f:

$$\frac{\partial f}{\partial t} = 0$$

$$\frac{\partial f}{\partial S} = \frac{1}{S_t}$$

$$\frac{\partial^2 f}{\partial S^2} = -\frac{1}{S_t^2}$$

Substituting these derivatives back into Itô's formula yields:

$$d(\ln(S_t)) = \frac{1}{S_t}dS_t + \frac{1}{2}\left(-\frac{1}{S_t^2}(dS_t)^2\right)$$

To evaluate the squared differential term $(dS_t)^2$, let's square our original SDE expression:

$$(dS_t)^2 = (\mu S_t dt + \sigma S_t dW_t)^2$$

$$(dS_t)^2 = \mu^2 S_t^2 (dt)^2 + 2\mu\sigma S_t^2 dt dW_t + \sigma^2 S_t^2 (dW_t)^2$$

Knowing the following properties:

- $dt.dt \to 0$
- $dt.dW_t \to 0$
- $dW_t.dW_t \to dt$

We finally get:

$$(dS_t)^2 = \sigma^2 S_t^2 dt$$

By substituting our original expression for dS_t and our newly calculated $(dS_t)^2$ into the equation, we have:

$$d(\ln(S_t)) = \frac{1}{S_t}(\mu S_t dt + \sigma S_t dW_T) - \frac{1}{2S_t^2}(\sigma^2 S_t^2 dt)$$

We distribute the terms across the fractions:

$$d(\ln(S_t)) = \mu dt + \sigma dW_t - \frac{1}{2}\sigma^2 dt$$

$$\Leftrightarrow d(\ln(S_t)) = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma dW_t$$

We can directly integrate both sides from the initial time ($t = 0$) to our final maturity horizon ($t = T$):

$$\int_0^T d(lnS_t) = \int_0^5 \left(\mu - \frac{\sigma^2}{2}\right) dt + \int_0^T \sigma dW_t$$

$$\Leftrightarrow \ln(S_T) - \ln(S_0) = \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma(W_T - W_0)$$

$$\Leftrightarrow \ln\left(\frac{S_T}{S_0}\right) = \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma W_T$$

$$\Leftrightarrow S_T = S_0 \ \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) T + \sigma W_T\right)$$

Appendix 3. — Student's t-distribution

We want to show that:

$$\lim_{\nu\to\infty} f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

. The PDF is given by:

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

If we analyze this into two independent parts, and using the fundamental definition of the exponential constant $\lim_{n\to+\infty}(1 + \frac{a}{n})^n = e^a$, we evaluate the limit of the right-hand term:

$$\lim_{\nu\to\infty} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}} = \lim_{\nu\to\infty} \left[(1 + \frac{x^2}{\nu})^{\nu}\right]^{-\frac{1}{2}} . \left(1 + \frac{x^2}{\nu}\right)^{-\frac{1}{2}}$$

Thus, the variable part yields exactly the Gaussian kernel: $e^{-\frac{x^2}{2}}$

Using the Striling's approximation for the Gamma function, which states that for large z:

$$\Gamma(z) \sim \sqrt{\frac{2\pi}{z}} \left(\frac{z}{e}\right)^z$$

Let's apply this to the ratio $\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})}$ by setting $z = \frac{\nu}{2}$:

$$\frac{\Gamma\left(z + \frac{1}{2}\right)}{\Gamma(z)} \sim \frac{\sqrt{\frac{2\pi}{z+1/2}} \left(\frac{z+1/2}{e}\right)^{z+1/2}}{\sqrt{\frac{2\pi}{z}} \left(\frac{z}{e}\right)^z}$$

$$= \sqrt{\frac{z}{z+1/2}} . e^{-1/2} . \frac{(z+1/2)^{z+1/2}}{z^z}$$

Rearranging the last fraction:

$$\frac{(z+1/2)^z . (z+1/2)^{1/2}}{z^z} = (1 + \frac{1}{2z})^z . \sqrt{z+1/2}$$

Taking the limit as $z \to \infty$:

- $\sqrt{\frac{z}{z+1/2}} \to 1$
- $\left(1 + \frac{1}{2z}\right)^z \to e^{1/2}$

Hence:

$$\lim_{z\to\infty} \frac{\Gamma(z + \frac{1}{2})}{\Gamma(z)} = \sqrt{z + \frac{1}{2}} \sim \sqrt{z} = \sqrt{\frac{\nu}{2}}$$

If we substitute this back into the normalization constant multiplier:

$$\lim_{\nu\to\infty} \frac{1}{\sqrt{\nu\pi}} \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} = \frac{1}{\sqrt{\nu\pi}} . \sqrt{\frac{\nu}{2}} = \frac{1}{\sqrt{2\pi}}$$

Combining both parts gives the standard normal density function:

$$f(x) \to \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Let T_ν be a standard Student's t-distributed random variable with ν degrees of freedom. By definition: $\mathbb{E}[T_\nu] = 0$ and $\mathbb{V}(T_\nu) = \frac{\nu}{\nu-2}$ for $\nu > 2$.

To scale a random variable so that its variance equals 1, we must divide by its standard deviation:

$$Z = \frac{T_\nu}{\sqrt{\mathbb{V}(T_\nu)}} = \frac{T_\nu}{\sqrt{\frac{\nu}{\nu-2}}} = T_\nu \sqrt{\frac{\nu-2}{\nu}}$$

Now Z represents a zero-mean, unit-variance process containing the heavy tails dictated by ν.

Under continuous geometric scaling over a time horizon T, with an expected annualized return μ and annualized volatility σ, the total return volatility scales by $\sqrt{T}$. We inject Z into the return dynamics:

$$\ln\left(\frac{S_T}{S_0}\right) = \mu T + \sigma\sqrt{T} Z$$

$$\Leftrightarrow \ln\left(\frac{S_T}{S_0}\right) = \mu T + \sigma\sqrt{T} \left(T_\nu \sqrt{\frac{\nu-2}{\nu}}\right)$$

$$\Leftrightarrow \ln\left(\frac{S_T}{S_0}\right) = \mu T + \sigma \sqrt{\frac{\nu-2}{\nu}} T_\nu \sqrt{T}$$

$$\Leftrightarrow S_T = S_0 \exp\left(\mu T + \sigma \sqrt{\frac{\nu-2}{\nu}} T_\nu \sqrt{T}\right)$$

Appendix 4. — Python code

```python
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Sun June 7 15:03:00 2026

@author: Quantitative Risk Management Framework
"""

import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt
import yfinance as yf

# Set global seed for exact reproducibility across all random generations
np.random.seed(42)

# ==========================================
# 1. MARKET DATA ACQUISITION & PORTFOLIO CONSTRUCTION
# ==========================================
print("FETCHING REAL-WORLD MARKET DATA VIA YFINANCE...")
```

```

# Define tickers for the requested global indices
# ^GSPC: S&P 500, ^IXIC: NASDAQ, ^FCHI: CAC 40, ^FTSE: FTSE 100, ^GDAXI: DAX, ^HSI: Hang Seng
tickers = {
    'S&P 500': '^GSPC',
    'NASDAQ': '^IXIC',
    'CAC 40': '^FCHI',
    'FTSE 100': '^FTSE',
    'DAX': '^GDAXI',
    'Hang Seng': '^HSI'
}

# Fetch 5 years of historical data to capture full market cycles/regimes
start_date = "2021-06-01"
end_date = "2026-06-01"

data = yf.download(list(tickers.values()), start=start_date, end=end_date)['Close']
data = data.dropna() # Drop non-overlapping holidays/weekends across global exchanges

# Compute daily log returns for statistical symmetry
asset_returns = np.log(data / data.shift(1)).dropna()

# Construct an equally-weighted portfolio
num_assets = len(tickers)
weights = np.array([1.0 / num_assets] * num_assets)

# Generate the historical portfolio returns time series
historical_returns = asset_returns.dot(weights).to_numpy()
n_days = len(historical_returns)

# Extract empirical parameters for log printout
emp_drift = np.mean(historical_returns)
emp_volatility = np.std(historical_returns)

# ==========================================
# 2. USER-DEFINED SIMULATION INPUT PARAMETERS
# ==========================================
M = 50000                   # Number of Monte Carlo scenarios
alpha = 0.975               # Confidence level for baseline CDF plotting (Plot 2)

initial_price = 100         # Starting asset price for trajectory paths simulation
n_path_steps = 100          # Horizon length (in days) to project the sample paths
n_paths_to_plot = 15        # Number of random paths to visualize per method

print("\nINITIALIZING PORTFOLIO STRESS TESTING ENGINE...")
print(f"Historical Sample Size: {n_days} trading days")
print(f"Empirical Daily Drift (Mean): {emp_drift:.4%}")
print(f"Empirical Daily Volatility (Std Dev): {emp_volatility:.4%}\n" + "-"*60)


# ==========================================
# APPROACH 1: GEOMETRIC BROWNIAN MOTION (NORMAL)
# ==========================================
# GBM extracts parameters via Maximum Likelihood Estimation, assuming perfect normality
mu_norm, sigma_norm = stats.norm.fit(historical_returns)

# 1A. Returns for VaR/CVaR static horizon
sim_returns_gbm = np.random.normal(loc=mu_norm, scale=sigma_norm, size=M)

# 1B. Dynamic path generation matrix (steps x paths)
paths_returns_gbm = np.random.normal(loc=mu_norm, scale=sigma_norm, size=(n_path_steps, n_paths_to_plot))
price_paths_gbm = initial_price * np.exp(np.vstack([np.zeros(n_paths_to_plot), np.cumsum(paths_returns_gbm, axis=0)]))


# ==========================================
# APPROACH 2: STUDENT'S T-DISTRIBUTION
# ==========================================
# Fits parameters directly to map history onto a theoretical t-curve (captures fat tails)
df_est, loc_est, scale_est = stats.t.fit(historical_returns)

# 2A. Returns for VaR/CVaR static horizon
sim_returns_t = stats.t.rvs(df=df_est, loc=loc_est, scale=scale_est, size=M)

# 2B. Dynamic path generation matrix
paths_returns_t = stats.t.rvs(df=df_est, loc=loc_est, scale=scale_est, size=(n_path_steps, n_paths_to_plot))
price_paths_t = initial_price * np.exp(np.vstack([np.zeros(n_paths_to_plot), np.cumsum(paths_returns_t, axis=0)]))


# ==========================================
# APPROACH 3: KERNEL DENSITY ESTIMATION (KDE)
# ==========================================
# Fits a non-parametric Gaussian KDE over the historical sample returns
kde = stats.gaussian_kde(historical_returns, bw_method='silverman')
h = kde.factor * np.std(historical_returns)

# 3A. Returns for VaR/CVaR static horizon (Hierarchical sampling)
random_indices = np.random.choice(n_days, size=M, replace=True)
x_I = historical_returns[random_indices]
Z = np.random.normal(0, 1, M)
sim_returns_kde = x_I + h * Z

# 3B. Dynamic path generation matrix via structured KDE drawing loops
paths_returns_kde = np.zeros((n_path_steps, n_paths_to_plot))
for step in range(n_path_steps):
    idx = np.random.choice(n_days, size=n_paths_to_plot, replace=True)
    paths_returns_kde[step, :] = historical_returns[idx] + h * np.random.normal(0, 1, n_paths_to_plot)
price_paths_kde = initial_price * np.exp(np.vstack([np.zeros(n_paths_to_plot), np.cumsum(paths_returns_kde, axis=0)]))


# ==========================================
# RISK METRICS EVALUATION FUNCTION
# ==========================================
def calculate_var_cvar(sim_returns, alpha=0.975):
    """
    Sorts returns and extracts Value at Risk and Conditional Value at Risk.
    Returns are transformed to positive numbers representing losses.
    """
    sorted_returns = np.sort(sim_returns)
    k = int(np.floor((1 - alpha) * len(sorted_returns)))

    var = -sorted_returns[k]
    cvar = -np.mean(sorted_returns[:k])
    return var, cvar


# ==========================================
# COMPUTE SYNCHRONIZED TARGET METRICS
# ==========================================
# Calculate specific cross-quantile targets for Plot 1 and Plot 3
var_gbm_99, _ = calculate_var_cvar(sim_returns_gbm, alpha=0.99)
var_t_99, _   = calculate_var_cvar(sim_returns_t, alpha=0.99)
var_kde_99, _ = calculate_var_cvar(sim_returns_kde, alpha=0.99)

_, cvar_gbm_975 = calculate_var_cvar(sim_returns_gbm, alpha=0.975)
_, cvar_t_975   = calculate_var_cvar(sim_returns_t, alpha=0.975)
_, cvar_kde_975 = calculate_var_cvar(sim_returns_kde, alpha=0.975)

# Calculate Skewness and Excess Kurtosis across frameworks
```

```python
skew_gbm, kurt_gbm = stats.skew(sim_returns_gbm), stats.kurtosis(sim_returns_gbm)
skew_t, kurt_t = stats.skew(sim_returns_t), stats.kurtosis(sim_returns_t)
skew_kde, kurt_kde = stats.skew(sim_returns_kde), stats.kurtosis(sim_returns_kde)


# ========================================
# STATISTICAL TESTS FOR SKEWNESS AND KURTOSIS
# ========================================
def test_skewness_kurtosis(data, name):
    """
    Perform statistical tests for significance of skewness and excess kurtosis.
    """
    n = len(data)
    sample_skew = stats.skew(data)
    sample_kurt = stats.kurtosis(data)  # Excess kurtosis

    if n < 4:
        return {
            'name': name, 'skew': sample_skew, 'kurt': sample_kurt,
            'skew_pvalue': np.nan, 'kurt_pvalue': np.nan, 'dagostino_pvalue': np.nan
        }

    se_skew = np.sqrt(6 * n * (n - 1) / ((n - 2) * (n + 1) * (n + 3)))
    se_kurt = np.sqrt(24 * n * (n - 1)**2 / ((n - 3) * (n - 2) * (n + 1) * (n + 3)))

    z_skew = sample_skew / se_skew if not np.isnan(se_skew) else np.nan
    z_kurt = sample_kurt / se_kurt if not np.isnan(se_kurt) else np.nan

    p_skew = 2 * (1 - stats.norm.cdf(abs(z_skew))) if not np.isnan(z_skew) else np.nan
    p_kurt = 2 * (1 - stats.norm.cdf(abs(z_kurt))) if not np.isnan(z_kurt) else np.nan

    dagostino_stat, dagostino_p = stats.normaltest(data)

    return {
        'name': name, 'skew': sample_skew, 'kurt': sample_kurt,
        'z_skew': z_skew, 'z_kurt': z_kurt,
        'skew_pvalue': p_skew, 'kurt_pvalue': p_kurt,
        'dagostino_stat': dagostino_stat, 'dagostino_pvalue': dagostino_p
    }


# Perform tests for each distribution
results_gbm = test_skewness_kurtosis(sim_returns_gbm, 'GBM (Normal)')
results_t = test_skewness_kurtosis(sim_returns_t, "Student's t")
results_kde = test_skewness_kurtosis(sim_returns_kde, 'KDE')


# ========================================
# RESULTS PRINT OUT
# ========================================
print(f"{'Approach':<30} | {'99.0% VaR':<10} | {'97.5% CVaR':<10} | {'Skewness':<9} | {'Ex. Kurt':<9}")
print("-" * 78)
print(f"{'1. Geometric Brownian (Normal)':<30} | {var_gbm_99:.4%}   | {cvar_gbm_975:.4%}   | {skew_gbm:+0.4f}  | {kurt_gbm:+0.4f}")
print(f"{'2. Student\'s t-Distribution':<30} | {var_t_99:.4%}   | {cvar_t_975:.4%}   | {skew_t:+0.4f}  | {kurt_t:+0.4f}")
print(f"{'3. Kernel Density Estimator':<30} | {var_kde_99:.4%}   | {cvar_kde_975:.4%}   | {skew_kde:+0.4f}  | {kurt_kde:+0.4f}")
print("-" * 78 + "\n")


# ========================================
# PRINT STATISTICAL TEST RESULTS
# ========================================
print("\n" + "=" * 80)
print("STATISTICAL SIGNIFICANCE TESTS FOR SKEWNESS AND KURTOSIS")
print("=" * 80)

for results in [results_gbm, results_t, results_kde]:
    skew_sig = '***' if results['skew_pvalue'] < 0.001 else '**' if results['skew_pvalue'] < 0.01 else '*' if results['skew_pvalue'] < 0.05 else ''
    kurt_sig = '***' if results['kurt_pvalue'] < 0.001 else '**' if results['kurt_pvalue'] < 0.01 else '*' if results['kurt_pvalue'] < 0.05 else ''
    dag_sig = '***' if results['dagostino_pvalue'] < 0.001 else '**' if results['dagostino_pvalue'] < 0.01 else '*' if results['dagostino_pvalue'] < 0.05 else ''

    print(f"\n{results['name']}:")
    print(f"  Skewness: {results['skew']:+0.4f} (z = {results['z_skew']:+0.4f}, p-value = {results['skew_pvalue']:.6f} {skew_sig})")
    print(f"  Ex. Kurtosis: {results['kurt']:+0.4f} (z = {results['z_kurt']:+0.4f}, p-value = {results['kurt_pvalue']:.6f} {kurt_sig})")
    print(f"  D'Agostino K^2: stat = {results['dagostino_stat']:+0.4f}, p-value = {results['dagostino_pvalue']:.6f} {dag_sig}")

print("\n" + "-" * 80)
print("Significance codes: '***' p < 0.001, '**' p < 0.01, '*' p < 0.05")
print("=" * 80 + "\n")

global_min_tail = np.percentile(historical_returns, 0.1)
global_max_tail = np.percentile(historical_returns, 15)

# Global styling mapping for the comparative charts
colors = {'gbm': '#1f77b4', 't': '#ff7f0e', 'kde': '#2ca02c'}

# ========================================
# PLOT 1: PROBABILITY DENSITY TAIL FOCUS
# ========================================
plt.figure(figsize=(10, 6))
plt.hist(sim_returns_gbm, bins=250, density=True, alpha=0.3, color=colors['gbm'], label='GBM (Normal)')
plt.hist(sim_returns_t, bins=250, density=True, alpha=0.3, color=colors['t'], label="Student's t")
plt.hist(sim_returns_kde, bins=250, density=True, alpha=0.3, color=colors['kde'], label='KDE')

plt.axvline(-var_gbm_99, color=colors['gbm'], linestyle=':', linewidth=2, label=f'GBM 99% VaR ({var_gbm_99:.2%})')
plt.axvline(-var_t_99, color=colors['t'], linestyle=':', linewidth=2, label=f'Student t 99% VaR ({var_t_99:.2%})')
plt.axvline(-var_kde_99, color=colors['kde'], linestyle=':', linewidth=2, label=f'KDE 99% VaR ({var_kde_99:.2%})')

plt.axvline(-cvar_gbm_975, color=colors['gbm'], linestyle='-', linewidth=2, label=f'GBM 97.5% CVaR ({cvar_gbm_975:.2%})')
plt.axvline(-cvar_t_975, color=colors['t'], linestyle='-', linewidth=2, label=f'Student t 97.5% CVaR ({cvar_t_975:.2%})')
plt.axvline(-cvar_kde_975, color=colors['kde'], linestyle='-', linewidth=2, label=f'KDE 97.5% CVaR ({cvar_kde_975:.2%})')

plt.title("Probability Density Focus: 99% VaR vs 97.5% CVaR Loss Tail (Real Data)", fontsize=13, fontweight='bold')
plt.xlabel("Daily Portfolio Returns")
plt.ylabel("Density")
plt.xlim(global_min_tail, global_max_tail)
plt.legend(fontsize=9, loc='upper left')
plt.grid(True, alpha=0.2)
plt.show()

# ========================================
# PLOT 2: CUMULATIVE DISTRIBUTION FUNCTION (CDF)
# ========================================
plt.figure(figsize=(10, 6))
```

```
for data, name, col in zip([sim_returns_gbm, sim_returns_t, sim_returns_kde],
                           ['GBM (Normal)', "Student's t", 'KDE'],
                           [colors['gbm'], colors['t'], colors['kde']]):
    sorted_data = np.sort(data)
    cdf = np.arange(1, len(sorted_data) + 1) / len(sorted_data)
    plt.plot(sorted_data, cdf, label=name, color=col, linewidth=2)

plt.axhline(1-0.975, color='red', linestyle='--', linewidth=1.2, label='Significance Level (0.025)')
plt.axhline(1-0.99, color='blue', linestyle='--', linewidth=1.2, label='Significance Level (0.01)')
plt.title("Empirical CDF Quantile Trajectory (Real Data)", fontsize=13, fontweight='bold')
plt.xlabel("Daily Portfolio Returns")
plt.ylabel("Cumulative Probability")
plt.xlim(np.percentile(historical_returns, 0.1), 0.01)
plt.ylim(0, 0.10)
plt.legend(fontsize=10, loc='lower right')
plt.grid(True, alpha=0.2)
plt.show()

# ========================================
# PLOT 3: DIRECT METRIC COMPARISON (BAR CHART)
# ========================================
fig, ax = plt.subplots(figsize=(10, 6))
categories = ['GBM (Normal)', "Student's t", 'KDE']
var_values = [var_gbm_99 * 100, var_t_99 * 100, var_kde_99 * 100]
cvar_values = [cvar_gbm_975 * 100, cvar_t_975 * 100, cvar_kde_975 * 100]

x = np.arange(len(categories))
width = 0.35
rects1 = ax.bar(x - width/2, var_values, width, label='99% VaR', color='#bcbd22', alpha=0.85)
rects2 = ax.bar(x + width/2, cvar_values, width, label='97.5% CVaR', color='#d62728', alpha=0.85)

ax.set_title("Direct Value Comparison: 99% VaR vs 97.5% CVaR (%)", fontsize=13, fontweight='bold')
ax.set_xticks(x)
ax.set_xticklabels(categories)
ax.set_ylabel("Loss Scale (%)")
ax.legend(fontsize=10, loc='upper left')
ax.grid(True, axis='y', alpha=0.3)

def autolabel(rects):
    for rect in rects:
        height = rect.get_height()
        ax.annotate(f'{height:.2f}%',
                    xy=(rect.get_x() + rect.get_width() / 2, height),
                    xytext=(0, 3), textcoords="offset points",
                    ha='center', va='bottom', fontsize=9, fontweight='bold')

autolabel(rects1)
autolabel(rects2)
plt.tight_layout()
plt.show()

# ========================================
# PLOTS 4, 5, 6: SIMULATED PRICE TRAJECTORIES
# ========================================
for paths, name, col in zip([price_paths_gbm, price_paths_t, price_paths_kde],
                            ['Geometric Brownian Motion (Normal)', "Student's t-Distribution", 'Kernel Density Estimator (KDE)'],
                            [colors['gbm'], colors['t'], colors['kde']]):
    plt.figure(figsize=(10, 4))
    plt.plot(paths, linewidth=1.5, color=col, alpha=0.6)
    plt.title(f"{name} Simulated Price Paths (Real Data)", fontsize=12, fontweight='bold')
    plt.xlabel("Time Horizon (Days)")
    plt.ylabel("Asset Price")
    plt.grid(True, alpha=0.3)
    plt.axhline(initial_price, color='black', linestyle='--', alpha=0.5)
    plt.show()

# ========================================
# PREPARATION FOR DISTRIBUTION OVERLAYS (PLOTS 7-9)
# ========================================
x_axis = np.linspace(np.min(historical_returns), np.max(historical_returns), 1000)
normal_curve = stats.norm.pdf(x_axis, loc=mu_norm, scale=sigma_norm)

# PLOT 7: HISTOGRAM - GBM
plt.figure(figsize=(10, 4))
plt.hist(sim_returns_gbm, bins=150, density=True, alpha=0.6, color=colors['gbm'], edgecolor='black', linewidth=0.5)
plt.plot(x_axis, normal_curve, color='red', linestyle='-', linewidth=2, label='Theoretical Normal Curve')
plt.title("Distribution of Returns - Geometric Brownian Motion (Normal)", fontsize=12, fontweight='bold')
plt.xlabel("Simulated Daily Returns")
plt.ylabel("Frequency Density")
plt.xlim(np.percentile(historical_returns, 0.1), np.percentile(historical_returns, 99.9))
plt.legend()
plt.grid(True, alpha=0.2)
plt.show()

# PLOT 8: HISTOGRAM - STUDENT'S T
plt.figure(figsize=(10, 4))
plt.hist(sim_returns_t, bins=150, density=True, alpha=0.6, color=colors['t'], edgecolor='black', linewidth=0.5)
plt.plot(x_axis, normal_curve, color='red', linestyle='-', linewidth=2, label='Theoretical Normal Curve')
t_curve = stats.t.pdf(x_axis, df=df_est, loc=loc_est, scale=scale_est)
plt.plot(x_axis, t_curve, color='black', linestyle='--', linewidth=2, label="Fitted Student's t Curve")
plt.title("Distribution of Returns - Student's t-Distribution", fontsize=12, fontweight='bold')
plt.xlabel("Simulated Daily Returns")
plt.ylabel("Frequency Density")
plt.xlim(np.percentile(historical_returns, 0.1), np.percentile(historical_returns, 99.9))
plt.legend()
plt.grid(True, alpha=0.2)
plt.show()

# PLOT 9: HISTOGRAM - KDE
plt.figure(figsize=(10, 4))
plt.hist(sim_returns_kde, bins=150, density=True, alpha=0.6, color=colors['kde'], edgecolor='black', linewidth=0.5)
plt.plot(x_axis, normal_curve, color='red', linestyle='-', linewidth=2, label='Theoretical Normal Curve')
plt.plot(x_axis, kde.pdf(x_axis), color='darkgreen', linestyle='-', linewidth=2, label='KDE Continuous Fit Line')
plt.title("Distribution of Returns - Kernel Density Estimator (KDE)", fontsize=12, fontweight='bold')
plt.xlabel("Simulated Daily Returns")
plt.ylabel("Frequency Density")
plt.xlim(np.percentile(historical_returns, 0.1), np.percentile(historical_returns, 99.9))
plt.legend()
plt.grid(True, alpha=0.2)
plt.show()
```

```
#!/usr/bin/env python3

import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt
import yfinance as yf
import warnings
warnings.filterwarnings('ignore')

np.random.seed(42)

# ========================================
# 1. Data Retrieval (Same portfolio as original paper)
# ========================================
print("Fetching data...")

```

```python
tickers = ['^GSPC', '^IXIC', '^FCHI', '^FTSE', '^GDAXI', '^HSI']
names = ['S&P500', 'NASDAQ', 'CAC40', 'FTSE100', 'DAX', 'HangSeng']

# Original paper: 2021-2026
# Split: 2015-2020 for training, 2021-2024 for validation
data_all = yf.download(tickers, start="2015-01-01", end="2024-12-31", auto_adjust=True)['Close']
data_all = data_all.dropna()

log_returns_all = np.log(data_all / data_all.shift(1)).dropna()
weights = np.array([1/6] * 6)
portfolio_returns_all = log_returns_all.dot(weights).to_numpy()
dates_all = log_returns_all.index

# Training period (for model fitting)
train_mask = log_returns_all.index < '2021-01-01'
train_returns = portfolio_returns_all[train_mask]

# Validation period (for backtesting)
test_mask = log_returns_all.index >= '2021-01-01'
test_returns = portfolio_returns_all[test_mask]
test_dates = dates_all[test_mask]

print(f"Training period: 2015-2020 ({len(train_returns)} days)")
print(f"Validation period: 2021-2024 ({len(test_returns)} days)")

# ==========================================
# 2. Model Fitting on Training Data
# ==========================================
M = 50000

# GBM
mu_norm, sigma_norm = stats.norm.fit(train_returns)
sim_gbm = np.random.normal(mu_norm, sigma_norm, M)

# Student's t
df_t, loc_t, scale_t = stats.t.fit(train_returns)
sim_t = stats.t.rvs(df=df_t, loc=loc_t, scale=scale_t, size=M)

# KDE
kde = stats.gaussian_kde(train_returns, bw_method='silverman')
h = kde.factor * np.std(train_returns)
idx = np.random.choice(len(train_returns), size=M, replace=True)
sim_kde = train_returns[idx] + h * np.random.normal(0, 1, M)

print(f"Student's t degrees of freedom: {df_t:.2f}")

# ==========================================
# 3. VaR/CVaR Calculation
# ==========================================
def calc_var_cvar(sim, alpha):
    sorted_r = np.sort(sim)
    k = int(np.floor((1 - alpha) * len(sorted_r)))
    var = -sorted_r[k]
    cvar = -np.mean(sorted_r[:k])
    return var, cvar

var_gbm, cvar_gbm = calc_var_cvar(sim_gbm, 0.99), calc_var_cvar(sim_gbm, 0.975)
var_t, cvar_t = calc_var_cvar(sim_t, 0.99), calc_var_cvar(sim_t, 0.975)
var_kde, cvar_kde = calc_var_cvar(sim_kde, 0.99), calc_var_cvar(sim_kde, 0.975)

var_gbm_99, cvar_gbm_975 = var_gbm[0], cvar_gbm[1]
var_t_99, cvar_t_975 = var_t[0], cvar_t[1]
var_kde_99, cvar_kde_975 = var_kde[0], cvar_kde[1]

print(f"\n{'Model':<15} {'VaR(99%)':<12} {'CVaR(97.5%)'}")
print("-"*40)
print(f"{'GBM':<15} {var_gbm_99:.4%}  {cvar_gbm_975:.4%}")
print(f"{'Student t':<15} {var_t_99:.4%}  {cvar_t_975:.4%}")
print(f"{'KDE':<15} {var_kde_99:.4%}  {cvar_kde_975:.4%}")

# ==========================================
# 4. Backtest (Validation period 2021-2024)
# ==========================================
print("\n=== Backtest Results (2021-2024) ===")
print("Theoretical: VaR(99%) violations should occur at 1% frequency\n")

viol_gbm = test_returns < -var_gbm_99
viol_t = test_returns < -var_t_99
viol_kde = test_returns < -var_kde_99

rate_gbm = np.mean(viol_gbm)
rate_t = np.mean(viol_t)
rate_kde = np.mean(viol_kde)

def judge(rate):
    if rate > 0.015:
        return " Underestimating risk (Dangerous)"
    elif rate < 0.005:
        return " Too conservative"
    else:
        return " Appropriate"

print(f"GBM violation rate:       {rate_gbm:.2%}  {judge(rate_gbm)}")
print(f"Student t violation rate: {rate_t:.2%}  {judge(rate_t)}")
print(f"KDE violation rate:       {rate_kde:.2%}  {judge(rate_kde)}")

# ==========================================
# 5. Major Event Validation
# ==========================================
print("\n=== Validation on Major Market Events ===")

events = {
    'COVID Crash (2020/3)': '2020-03-16',
    'Russia Invasion (2022/2)': '2022-02-24',
    'Fed Rate Shock (2022/6)': '2022-06-13',
    'SVB Collapse (2023/3)': '2023-03-13',
}

port_series = log_returns_all.dot(weights)

for event_name, date in events.items():
    try:
        nearby = port_series[date:date]
        if len(nearby) == 0:
            nearby = port_series.iloc[port_series.index.get_indexer([date], method='nearest')]
        val = nearby.iloc[0]
        gbm_breach = " Exceeded" if val < -var_gbm_99 else "OK"
        t_breach = " Exceeded" if val < -var_t_99 else "OK"
        kde_breach = " Exceeded" if val < -var_kde_99 else "OK"
        print(f"{event_name}: {val:.2%} | GBM:{gbm_breach} | t:{t_breach} | KDE:{kde_breach}")
    except Exception as e:
        print(f"{event_name}: No data available")

# ==========================================
# 6. Visualization
# ==========================================
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('VaR/CVaR Backtest (Train:2015-2020, Test:2021-2024)', fontsize=13, fontweight='bold')

# Return distribution
ax1 = axes[0, 0]
ax1.hist(train_returns, bins=100, density=True, alpha=0.5, color='gray', label='Actual (Train)')
x = np.linspace(train_returns.min(), train_returns.max(), 1000)
ax1.plot(x, stats.norm.pdf(x, mu_norm, sigma_norm), 'b-', lw=2, label='GBM')
ax1.plot(x, stats.t.pdf(x, df_t, loc_t, scale_t), 'r-', lw=2, label="Student t")
ax1.plot(x, kde.pdf(x), 'g-', lw=2, label='KDE')
ax1.set_title('Return Distribution (Training Period)')
ax1.legend(fontsize=8)
ax1.set_xlim(-0.08, 0.06)
```

```python

# VaR/CVaR bar chart
ax2 = axes[0, 1]
models = ['GBM', "Student t", 'KDE']
vars_val = [var_gbm_99*100, var_t_99*100, var_kde_99*100]
cvars_val = [cvar_gbm_975*100, cvar_t_975*100,
↪  cvar_kde_975*100]
x_pos = np.arange(3)
ax2.bar(x_pos - 0.2, vars_val, 0.4, label='VaR(99%)',
↪  color='steelblue', alpha=0.8)
ax2.bar(x_pos + 0.2, cvars_val, 0.4, label='CVaR(97.5%)',
↪  color='tomato', alpha=0.8)
ax2.set_xticks(x_pos)
ax2.set_xticklabels(models)
ax2.set_ylabel('Loss Rate (%)')
ax2.set_title('VaR/CVaR Comparison')
ax2.legend()
for i, (v, c) in enumerate(zip(vars_val, cvars_val)):
    ax2.text(i-0.2, v+0.01, f'{v:.2f}%', ha='center',
    ↪  fontsize=8)
    ax2.text(i+0.2, c+0.01, f'{c:.2f}%', ha='center',
    ↪  fontsize=8)

# Time series with VaR lines
ax3 = axes[1, 0]
ax3.plot(test_dates, test_returns, color='gray',
↪  alpha=0.5, lw=0.5, label='Returns (Validation)')
ax3.axhline(-var_gbm_99, color='blue', lw=1.5,
↪  linestyle='--', label=f'GBM VaR {var_gbm_99:.2%}')
ax3.axhline(-var_t_99, color='red', lw=1.5,
↪  linestyle='--', label=f"t VaR {var_t_99:.2%}")
ax3.axhline(-var_kde_99, color='green', lw=1.5,
↪  linestyle='--', label=f'KDE VaR {var_kde_99:.2%}')
breach_dates = test_dates[viol_gbm]
ax3.scatter(breach_dates, test_returns[viol_gbm],
↪  color='blue', s=10, alpha=0.7, zorder=5)
ax3.set_title('Returns and VaR Violations (Validation
↪  Period)')
ax3.legend(fontsize=7)

# Violation rate bar chart
ax4 = axes[1, 1]
rates = [rate_gbm*100, rate_t*100, rate_kde*100]
colors_b = ['steelblue', 'tomato', 'green']
bars = ax4.bar(models, rates, color=colors_b, alpha=0.8)
ax4.axhline(1.0, color='black', lw=2, linestyle='-',
↪  label='Theoretical 1.00%')
ax4.set_ylabel('Violation Rate (%)')
ax4.set_title('VaR(99%) Violation Rate (Validation
↪  Period)')
ax4.legend()
for bar, rate in zip(bars, rates):
    ax4.text(bar.get_x() + bar.get_width()/2,
    ↪  bar.get_height() + 0.02,
             f'{rate:.2f}%', ha='center', fontsize=10,
             ↪  fontweight='bold')

plt.tight_layout()
plt.savefig('var_backtest_result.png', dpi=150,
↪  bbox_inches='tight')
plt.show()
print("\nDone. Saved as var_backtest_result.png")

```